Twelve Integer Variables through Interpolated Pell Certificates

A monotone intersective merger, trace before separation,
and explicit Diophantine equations for syntactic consistency

Research contribution prepared for
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Abstract

We construct an ordinary polynomial in twelve unrestricted integer variables, with exact total degree 4832 and 248 binary arithmetic operations, whose lack of integer solutions is equivalent in ZFC to syntactic Con(ZFC), by composition with explicitly named ordinary calibration theorems already used in ProveIt. The new reduction shares a positivity witness with a strong Pell certificate. Full residue interpolation of the auxiliary Pell coordinate then lets that same coordinate enforce a divisibility condition and exclude a zero multiplier. A strictly increasing intersective quintic, $(x^2+31)(x^3+x+1)$, performs this merger while the main Pell square root is still algebraic. Its absence of roots in real quadratic fields and the strict positivity of its divided difference are essential.

A second construction uses thirteen unrestricted integers, exact degree 1072, and 190 binary arithmetic operations, improving the inspected thirteen-integer 1088-degree/195-operation construction in both measures. Its new ingredient is a normalized field trace that recovers a congruence before square-root separation. Both polynomials have fully specified fixed coefficients, literal division-free arithmetic graphs, and offline checks. We make no claim of global minimality, a world variable-count record, or historical priority for every constituent idea. The new proofs are ordinary mathematical proofs; this package does not add an end-to-end Lean or encoded object-language ZFC derivation.

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1 What is improved, and what is meant by an equation

The task is to specify a polynomial $P \in \mathbb{Z}[x_1, \ldots, x_m]$, together with its domain, for which

$$\mathrm{ZFC} \vdash \left[\neg\exists \mathbf{x} \in \mathbb{Z}^m \; P(\mathbf{x}) = 0\right] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1.1}$$

The consistency sentence refers to the recursive finite-proof coding in the inherited calibration. Proving this biconditional does not prove either side of it. No coefficient is selected by deciding consistency, waiting for a proof search, or choosing an unspecified enumeration index.

There is no single numerical meaning of "simplest." A polynomial with fewer variables may have much larger degree, a shorter circuit may use large constants, and an expanded expression may be enormous despite a small circuit. We preserve these distinctions. An arithmetic circuit here has only integer leaves, the declared variable leaves, and binary addition, subtraction and multiplication. Squaring costs a multiplication; fixed-scalar multiplication is charged; reuse of a computed value is free. Every intermediate letter in a formula below is a polynomial abbreviation, not an additional quantified coordinate.

The comparison snapshot includes the requested Hilbert Tenth directory, the Diophantine consistency collection, and all fifteen ZIP archives then present in `docs/incoming`. Their hashes are recorded in the package. The incoming root-fusion manuscript already contains a thirteen-integer construction of degree 1088 and 195 operations. The active development and other incoming proposals include ten-integer constructions of much larger degree. Thus the twelve-variable result is a new tradeoff within the inspected construction families, not a new universal variable bound. Sun's recent ten-unknown representation theorem also rules out presenting twelve unknowns as the current general record [9].

Construction	Vars.	Degree	Arithmetic
Incoming five-root CRT norm	13	1088	195 operations
Trace-first companion, this package	13	1072	190 operations
Interpolated quintic merger, this package	12	4832	248 operations
Shorter twelve-variable companion	12	16000	223 operations

Table 1: The directly relevant comparison. All domains are unrestricted integers. The twelve-variable tradeoff spends degree to remove one more coordinate. The machine-readable metrics report its full operation census.

The main conceptual advance is not merely an arithmetic rewrite. The former construction represents a strict positivity condition using three square/triangular coordinates, one of which is absorbed into a norm. The new construction uses a Pell positivity witness, shares its multiplier with the strong certificate, and reuses the auxiliary coordinate as an intersective residue. The proof must establish full residue coverage; ordinary existence of Pell solutions would not suffice.

2 Fixed data and the calibrated arithmetic interface

Both constructions use the same coherent six-integer tuple:

$$\beta = 2^{1064921} - 1, \quad K_0, \quad M_C, \quad M_F, \quad A_c = 2^{1113}, \quad B_c = 2^{666104}. \tag{2.1}$$

The full finite recipe and all actual numeral bytes are included. In particular $M_F = M_{F,\mathrm{native}} + \beta$ is already shifted. It must not be shifted again. Appendix A prints the bounded recipe. The ordered 943-tile table and 430 machine actions are supplied data, so the recipe specifies actual integers rather than free parameters.

For the twelve-variable construction use precisely

$$\boxed{j, C, z, h, r, v, w, T, c, n, t, s \in \mathbb{Z}.} \tag{2.2}$$

Capital C and lowercase c are distinct coordinates. Define

$$\begin{aligned} J &= 6j+1, & q &= \beta J + 1, & W_0 &= 12w + 4, \\ X &= qW_0, & Y &= (2v+1)q^3, & E &= XY, \\ a &= E + Y, & A &= a + 2, & H &= 4a + 3, & \Delta &= A^2 - 1 = a^2 + H. \end{aligned} \tag{2.3}$$

Use the polynomial transport substitution

$$\begin{aligned} F &= (K_0 + W_0)C - z(q-1), & \alpha &= q - F - C, \\ R &= (q(q-F) - C)(q^2-1) + (M_C + qM_F)J, \\ k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta, \\ b &= X + ac, & D_{\mathrm{m}} &= 1 + \Delta c^2, \\ L_\tau &= XY^2 k, & D_\tau &= 1 + L_\tau(L_\tau + k). \end{aligned} \tag{2.4}$$

The last factorization is the established identity $XY^2(XY^2+1)k^2 = L_\tau(L_\tau + k)$; its reuse is credited to the existing Hilbert Tenth construction.

The two strict guards are combined by

$$\begin{aligned} U_c &= A_c F, & Z_c &= B_c C, \\ G &= U_c Z_c - (q - U_c - Z_c)^2, & P_\eta &= \eta\zeta, \\ O &= GP_\eta - (t^2 - G - P_\eta + 1)^2. \end{aligned} \tag{2.5}$$

For integer x, y, there is an integer t with

$$\mathrm{Pair}(x, y, t) := xy - (t^2 - x - y + 1)^2 > 0 \quad \Longleftrightarrow \quad x > 0 \text{ and } y > 0. \tag{2.6}$$

The forward implication follows because two negative factors make the squared term larger than their product. For the converse put $N = x + y - 1$ and choose the nearest integer square t^2. If $m = \lfloor \sqrt{N} \rfloor$, one of $m, m+1$ gives $|t^2 - N| \le m$, with strict squared error $< N \le xy$ unless the error is zero. This proves the claim. The square root in this witness selection is not an operation in the Diophantine polynomial.

The inherited signed Pell/native interface identifies finite ZFC refutations with canonical solutions of the transport, first and main Pell equations, the strict guards, and a strong/auxiliary rank certificate. In the completeness direction it provides

$$A > 2 \text{ even}, \quad R > 0, \quad R \equiv 3 \pmod 4, \quad c = \psi_A(R) > 0, \quad d = \chi_A(R) = b + H\gamma, \tag{2.7}$$

with integral γ, together with $\tau^2 = D_\tau$ and $G, P_\eta > 0$. Here

$$\chi_A(m) + \psi_A(m)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^m. \tag{2.8}$$

The determinant guard uses canonical native witnesses; it is an existence-preserving restriction, not a statement about every older witness.

Our strong square is already normalized: $f^2 - \Delta I^2 = 1$ with $I = src^2$. A direct rank argument in Section 9 therefore supplies the strong-index conclusion without extending an odd-only source parametrization. The auxiliary congruences remain $V \equiv -c \pmod f$ and $V \equiv -R \pmod c$. The new merger proves $I \neq 0$. No half-integer old coordinate is introduced.

3 A monotone intersective quintic

Define

$$\Pi(x) = (x^2 + 31)(x^3 + x + 1) = x^5 + 32x^3 + x^2 + 31x + 31. \tag{3.1}$$

The particular combination of properties below is what the construction uses. The general technique of intersective divisibility mergers is classical and is used in Sun's recent reduction; the inspected repository also already uses the different quintic $(x^2 + 23)(x^3 - x - 1)$.

Lemma 3.1. *The polynomial Π has a root modulo every positive integer, has no root in any real extension of $\mathbb{Q}$ of degree at most two, and is strictly increasing on $\mathbb{R}$.*

Proof. The cubic $f = x^3 + x + 1$ has discriminant -31 and no rational root: the only possible rational roots are $1, -1$, and neither is a root. It is therefore irreducible over $\mathbb{Q}$. Its roots cannot lie in a field of degree at most two. The quadratic $x^2 + 31$ has no real root.

For an odd prime $p \neq 31$, either f has a root in $\mathcal{F}_p$, or it is an irreducible cubic. In the latter case Frobenius permutes its three roots as a three-cycle. The Vandermonde product is fixed by that even permutation, so its square, the discriminant -31, is a nonzero square in $\mathcal{F}_p$. Thus $x^2 + 31$ has a simple root. In the former case a cubic root is simple because the discriminant is nonzero modulo p. At $p = 31$, $x = 3$ is a simple cubic root, since $f(3) = 31$ and $f'(3) = 28$. These simple roots lift to every power of the corresponding odd prime. At 2, $-31 \equiv 1 \pmod 8$, and an odd square root lifts successively to every 2-power. The Chinese remainder theorem supplies a root modulo any positive integer.

Finally

$$\Pi'(x) = 5x^4 + 96x^2 + 2x + 31 = 5x^4 + 95x^2 + (x + 1)^2 + 30 > 0.$$

□

For nonzero integer H put

$$\Theta(H, \xi, T) = H^5 \Pi\left(\frac{\xi + HT}{H}\right). \tag{3.2}$$

This is a homogeneous integer polynomial of degree five in $H, \xi + HT$. If ξ is integral and $H^5 \mid \Theta(H, \xi, T)$, then $H \mid \xi$. Indeed, if p has $\nu_p(\xi) < \nu_p(H)$, the monic leading term $(\xi + HT)^5$ has strictly smaller p-valuation than every other term. Its valuation is $5\nu_p(\xi) < 5\nu_p(H)$, a contradiction.

The merger must operate before the main root $d = \sqrt{D_{\mathrm{m}}}$ is known to be integral. Set $u = HT - b$. Reducing $d^2 = D_{\mathrm{m}}$ gives

$$\Theta(H, d - b, T) = \mathcal{A} + \mathcal{B}d, \tag{3.3}$$

where

$$\begin{aligned} \mathcal{A} &= u^5 + 10u^3 D_{\mathrm{m}} + 5uD_{\mathrm{m}}^2 + 32H^2(u^3 + 3uD_{\mathrm{m}}) + H^3(u^2 + D_{\mathrm{m}}) + 31H^4 u + 31H^5, \\ \mathcal{B} &= 5u^4 + 10u^2 D_{\mathrm{m}} + D_{\mathrm{m}}^2 + 32H^2(3u^2 + D_{\mathrm{m}}) + 2H^3 u + 31H^4. \end{aligned} \tag{3.4}$$

Since $D_{\mathrm{m}} > 0$ and $H \neq 0$, the coefficient $\mathcal{B}$ is positive. One purely algebraic certificate is

$$\mathcal{B} = 5u^4 + 10u^2 D_{\mathrm{m}} + D_{\mathrm{m}}^2 + 32H^2 D_{\mathrm{m}} + 95H^2 u^2 + H^2(u + H)^2 + 30H^4 > 0.$$

Also $\mathcal{A} \pm \mathcal{B}\sqrt{D_{\mathrm{m}}} \neq 0$, by Lemma 3.1, even if D_{m} is not a square. This excludes a zero multiplier without first assuming squarehood.

4 The twelve-variable polynomial

Keep all definitions in Section 2 and define

$$\begin{aligned} I &= src^2, & D_{\mathrm{f}} &= 1+\Delta I^2, & Q &= \Delta^2 I^2,\\ V &= D_{\mathrm{f}}(c-R+4IT)-c, & D_{\mathrm{a}} &= (Q-1)(QV^2-1), & & \\ D_{\mathrm{g}} &= (3O-4)s^2+1, & S &= IH^5, & Z &= Sn-\mathcal{A}. \end{aligned} \tag{4.1}$$

The two uses of T are intentional: it is both the auxiliary Pell coordinate in V and the intersective residue in $u = HT - b$. Put $\ell_0 = XY^2$. Use bounds on the root magnitudes, with one bound for each stage:

$$\begin{aligned} K_T &= T^2+c^2+1, & B_0 &= 256K_T^2, & & \\ B_b &= 1+c^2+R+c^2(s^2+r^2)(T^2+1), & & & & \\ B_{\mathrm{a}} &= Q(D_{\mathrm{f}}B_b+c^2+1), & B_{\mathrm{f}} &= \Delta+I^2+3, & B_\tau &= 2+\ell_0^2+k^2. \end{aligned} \tag{4.2}$$

The four separating weights are

$$\begin{aligned} W_1 &= 2B_0+1, & W_2 &= W_1(2B_{\mathrm{a}}+1),\\ W_3 &= W_2(2B_{\mathrm{f}}+1), & W_4 &= W_3(2B_\tau+1). \end{aligned} \tag{4.3}$$

They are odd, hence nonzero, on every integer assignment. Their positivity will be proved from a vanishing norm factor before using them to separate roots. Define the five integer polynomial radicands

$$\begin{aligned} T_0 &= \mathcal{B}^2 D_{\mathrm{m}}, & T_1 &= S^2W_1^2D_{\mathrm{a}}, & & \\ T_2 &= S^2W_2^2D_{\mathrm{f}}, & T_3 &= S^2W_3^2D_\tau, & T_4 &= S^2W_4^2D_{\mathrm{g}}. \end{aligned} \tag{4.4}$$

For formal radicals let

$$\mathcal{F}_{32}(z;t_0,t_1,t_2,t_3,t_4) = \prod_{\varepsilon\in\{-1,1\}^5}\left(z-\sum_{i=0}^{4}\varepsilon_i\sqrt{t_i}\right). \tag{4.5}$$

The sign symmetries make this an element of $\mathbb{Z}[z,t_0,\dots,t_4]$. The requested equation is

$$\boxed{P_{12}(j,C,z,h,r,v,w,T,c,n,t,s) := \mathcal{F}_{32}(Z;T_0,T_1,T_2,T_3,T_4) = 0.} \tag{4.6}$$

There is no division in this formula and no variable exponent. The radical product defines a finite polynomial. The delivered literal graph implements it using only $+,-,\times$; a small recurrence is supplied in Section 7.

5 Soundness of the twelve-variable reduction

Lemma 5.1 (Adjacent-coefficient descent)**.** *For integers $Q>1, V$ and rational e, the equation $QV^2-(Q-1)e^2=1$ implies $e\in\mathbb{Z}$.*

Proof. Write $e=a/d$ in lowest terms. Then $d^2 \mid Q-1$. If $|V|=1$, then $|e|=1$. Otherwise take positive absolute values and set

$$V' = (2Q-1)V-2(Q-1)e, \qquad e' = (2Q-1)e-2QV.$$

The norm is unchanged; V' is integral; and the reduced denominator of e' remains d, since $2Q-1\equiv 1 \pmod d$. The inequalities $e>V$ and $(Q-1)e<QV$ give $-V<V'<V$. The norm excludes $V'=0$. Descent in the positive integer $|V|$ reaches 1 without changing the denominator, which must consequently have been 1. □

Lemma 5.2 (Separation over a multiquadratic field)**.** *If nonzero complex square roots of rational numbers have strictly superincreasing absolute values and their signed sum is rational, every one of those roots is rational.*

Proof. If a root is irrational, some automorphism of their multiquadratic splitting field changes its sign. Subtracting the conjugate rational relation yields a nonempty vanishing signed subsum. Its largest term has absolute value greater than the sum of the absolute values of the others, contradicting the triangle inequality. Square classes need not be independent, and negative radicands are allowed. □

For every integer assignment to the front end, q is even, $q \equiv 2 \pmod 3$, $|q| \geq 16$, $|X| \geq 64$ and $|Y| \geq 4096$. Thus $|a| \geq 63 \cdot 4096$, $H \neq 0$ and $\Delta > 1$. The radicands $D_{\mathrm{m}}, D_{\mathrm{f}}, D_\tau$ are positive: for the last use $u(u+1) \geq 0$ for the integer $u = XY^2$.

Suppose $P_{12} = 0$. If $S = 0$, the four radicands $T_1, \ldots, T_4$ are zero and the polynomial specializes to

$$(\mathcal{A}^2 - \mathcal{B}^2 D_{\mathrm{m}})^{16}.$$

Both factors $\mathcal{A} \pm \mathcal{B}\sqrt{D_{\mathrm{m}}}$ are nonzero by Lemma 3.1. This is a contradiction. Therefore

$$S \neq 0, \qquad I \neq 0, \qquad s \neq 0, \quad r \neq 0, \quad c \neq 0. \tag{5.1}$$

In particular $Q > 1$ and $D_{\mathrm{f}} = 1 + \Delta s^2 r^2 c^4 > |c| + 1$. Writing $b' = c - R + 4IT \in \mathbb{Z}$, either $b' = 0$ and $|V| = |c| \geq 1$, or $|V| \geq D_{\mathrm{f}} - |c| > 1$. Hence $D_{\mathrm{a}} > 0$.

We first recover positivity without assuming any individual root is rational. The weights W_i are odd, hence nonzero. All radicands except D_{g} are positive. If $D_{\mathrm{g}} < 0$, its radical would be the only nonzero imaginary term in a vanishing factor, which is impossible. Thus $D_{\mathrm{g}} \geq 0$. Since $s \neq 0$, its defining expression forces $O > 0$: if $O \leq 0$, then $D_{\mathrm{g}} \leq 1 - 4s^2 < 0$. The pairing lemma gives $G > 0$ and $\eta\zeta > 0$.

We also need $R > 0$ before bounding the roots. The determinant inequality implies that F, C, α have the sign of q. To see this, U_c, Z_c have the same sign and $|U_c + Z_c| < 2|q|$; since $A_c, B_c \geq 2$, we get $|F| + |C| < |q|$. If $q > 0$, then $q(q - F) - C = q\alpha + (q-1)C > 0$, and the mask term in R is positive. If $q = -q_1 < 0$, write $C = -C_1$, $\alpha = -\alpha_1$ with $C_1, \alpha_1 > 0$. Then $q(q - F) - C = q_1(C_1 + \alpha_1) + C_1 > 0$. Also $J < 0$ and $M_C + qM_F = M_C - q_1 M_F < 0$, because the actual fixed tuple has $M_F > \beta > M_C > 0$. Thus $R > 0$ in this case as well.

The three bounds $B_{\mathrm{a}}, B_{\mathrm{f}}, B_\tau$ are now positive. The inequality

$$4|srT| \leq (s^2 + r^2)(T^2 + 1)$$

gives $|c - R + 4IT| \leq B_b$ and $|V| \leq D_{\mathrm{f}} B_b + c^2 + 1$. Consequently

$$\sqrt{D_{\mathrm{a}}} < Q|V| \leq B_{\mathrm{a}}, \qquad \sqrt{D_{\mathrm{f}}} \leq B_{\mathrm{f}}, \qquad \sqrt{D_\tau} \leq B_\tau. \tag{5.2}$$

For the last two, use $\sqrt{1 + \Delta I^2} \leq 1 + (\Delta + I^2)/2$ and $\sqrt{\ell_0(\ell_0 + 1)} \leq |\ell_0| + 1$, together with $2|xy| \leq x^2 + y^2$.

The main radical admits a much smaller bound than its polynomial numerator suggests. Put $v_0 = u/H$ and $z_0 = \sqrt{D_{\mathrm{m}}}/|H|$. The uniform front-end size bounds imply

$$\left|\frac{b}{H}\right| < 1 + \frac{|c|}{3}, \qquad z_0 < \frac{|c|}{2}.$$

Indeed $|a/H| < 1/3$, $|X/H| < 1$ and $\sqrt{1 + \Delta c^2} < (|a| + 3)|c|$, while $|H| \geq 4|a| - 3 > 2(|a| + 3)$. Thus $v_0^2 \leq 3K_T$ and $z_0^2 \leq K_T$. Homogeneity of the quintic gives

$$\begin{aligned} \frac{\mathcal{B}}{H^4} &= 5v_0^4 + 10v_0^2 z_0^2 + z_0^4 + 96v_0^2 + 32z_0^2 + 2v_0 + 31 \\ &\leq 76K_T^2 + 323K_T + 32 \leq 431K_T^2. \end{aligned}$$

Since $|I| = |sr|c^2 \geq c^2$,

$$\frac{\mathcal{B}\sqrt{D_{\mathrm{m}}}}{|S|} = \frac{\mathcal{B}}{H^4}\frac{z_0}{|I|} < \frac{431}{2}K_T^2 < 256K_T^2 = B_0. \tag{5.3}$$

This uses the denominator already present in the congruence merger; it adds no coordinate and no division to the polynomial.

Divide a vanishing norm factor by S. The five magnitudes, in order, are

$$\frac{\mathcal{B}\sqrt{D_{\mathrm{m}}}}{|S|}, \quad W_1\sqrt{D_{\mathrm{a}}}, \quad W_2\sqrt{D_{\mathrm{f}}}, \quad W_3\sqrt{D_\tau}, \quad W_4\sqrt{D_{\mathrm{g}}}.$$

The first is less than $W_1/2$. After adding the auxiliary term, the sum of the first two is less than $W_1(1/2 + B_{\mathrm{a}}) = W_2/2$. The same induction bounds the sum of the first three by $W_3/2$ and the first four by $W_4/2$. Every nonzero root of an integer radicand has magnitude at least 1. Hence these terms are strictly superincreasing, after a zero guard term is omitted. No upper bound on the last guard radical is needed. Lemma 5.2 makes all five roots rational. Their integer radicands and $\mathcal{B} \neq 0$ give signed integers

$$d^2 = D_{\mathrm{m}}, \quad a_1^2 = D_{\mathrm{a}}, \quad g^2 = D_{\mathrm{g}}, \quad \tau^2 = D_\tau, \quad f^2 = D_{\mathrm{f}}.$$

By Lemma 5.1, $e = a_1/(Q-1)$ is integral and $QV^2 - (Q-1)e^2 = 1$.

The integral factor relation now implies

$$S \mid \mathcal{A} + \mathcal{B}d = \Theta(H, d-b, T). \tag{5.4}$$

The monic valuation argument following (3.2) gives $H \mid d-b$, so $\gamma = (d-b)/H$ is an integer and the main congruence is restored. The previously recovered guards $G > 0$ and $\eta\zeta > 0$ remain available.

Set $\rho = \Delta I$. Then

$$\rho^2 = \Delta(f^2-1), \qquad c^2 \mid \rho, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}.$$

For the last congruence use $c^2 \mid I$ and $D_{\mathrm{f}} \equiv 1 \pmod{c}$. The direct normalized rank argument in Section 9 now recovers the signed native interface. Consequently a polynomial zero gives a finite refutation under the inherited native calibration.

6 Full residue interpolation and completeness

The crucial converse allows T to take every residue modulo a fixed strong witness I. Choosing a separate I after choosing each residue would not prove the statement needed by the merger.

Lemma 6.1 (Positivity by a Pell square)**.** *For every integer O,*

$$O > 0 \iff \exists s \in \mathbb{Z} \setminus \{0\}\ [(3O-4)s^2 + 1 \text{ is a square}].$$

Proof. For $O \leq 0$ the expression is negative. At $O = 1$, take $s = 1$, giving zero. For $O \geq 2$, the positive integer $3O - 4$ is 2 modulo 3, hence nonsquare. The ordinary Pell equation supplies a nontrivial solution. This is the positivity device used in Sun's relation-combining argument [9]. □

Theorem 6.2 (Native auxiliary interpolation)**.** *Let $A > 2$ be even, $\Delta = A^2 - 1$, $R > 0$ with $R \equiv 3 \pmod 4$, and $c = \psi_A(R)$. Given any positive integer N, there is a positive integer I divisible by Nc^2 and an integer $f > 0$ satisfying $f^2 - \Delta I^2 = 1$, such that for every residue $a_0 \pmod I$ there is an integer T for which*

$$V = f^2(c - R + 4IT) - c, \qquad (\Delta I)^2V^2 - ((\Delta I)^2 - 1)e^2 = 1$$

has an integer e. The chosen T can be made arbitrarily large and satisfies $T \equiv a_0 \pmod I$.

Proof. Finite order of the Pell matrix modulo $4Nc^2$ gives an index $m > 0$ with $I = \psi_A(m)$ divisible by $4Nc^2$. Put $f = \chi_A(m)$. In particular m is even, f is odd and $\gcd(f, I) = 1$. The standard Lucas valuation formulas give

$$p \mid I \quad \Longrightarrow \quad \nu_p(m) \le \nu_p(I). \tag{6.1}$$

For odd primes these are the rank-of-apparition valuation formulas; at 2, since A is even, $\nu_2(\psi_A(m)) = \nu_2(A) + \nu_2(m)$. These are exactly the specializations of the formulas used in the full-residue argument of Sun [9, 10]. It follows that

$$\gcd(4mf, 4I^2) \mid 4I.$$

The factor f causes no obstruction because it is coprime to I.

Choose an arbitrarily large positive ℓ by the Chinese remainder theorem with

$$\ell \equiv R \pmod{4mf}, \qquad \ell \equiv R - 4Ia_0 \pmod{4I^2}. \tag{6.2}$$

In particular $\ell \equiv 3 \pmod 4$. The period $4mf$, rather than merely $4m$, is necessary modulo f^2. Indeed for $\alpha = A + \sqrt{\Delta}$ we have $\alpha^{2m} = 2f\alpha^m - 1$. Squaring and then raising to the fth power gives $\alpha^{4mf} \equiv 1 \pmod{f^2}$ in $\mathbb{Z}[\sqrt{\Delta}]$.

Set $D = \Delta I$ and

$$V_0 = \chi_D(\ell)/D, \qquad e = \psi_D(\ell).$$

For odd ℓ, $\chi_D(\ell)$ is divisible by D, and the Pell identity gives $D^2V_0^2 - (D^2 - 1)e^2 = 1$. The odd Chebyshev identity, using $D^2 \equiv -\Delta \pmod{f^2}$, gives

$$V_0 \equiv -\psi_A(\ell) \equiv -c \pmod{f^2}.$$

For $\ell \equiv 3 \pmod 4$, the same odd polynomial has constant term $-\ell$ and every nonconstant term is divisible by $4D^2$; hence

$$V_0 \equiv -\ell \pmod{4I^2}.$$

Since A is even, the Pell recurrence modulo 4 gives $c = \psi_A(R) \equiv R \pmod 4$. Also $f^2 = 1 + \Delta I^2$, so

$$V_0 + c - f^2(c - R) \equiv R - \ell \equiv 4Ia_0 \pmod{4I^2}.$$

This numerator is divisible by f^2 and by $4I$. As f is odd and coprime to I, define the integer

$$T = \frac{V_0 + c - f^2(c - R)}{4If^2}.$$

Dividing the preceding congruence by $4I$, and using $f^2 \equiv 1 \pmod I$, gives $T \equiv a_0 \pmod I$. As ℓ tends to infinity through the CRT progression, V_0 and then T tend to infinity. This proves full residue coverage for the single chosen pair (I, f). □

Theorem 6.3 (Completeness of the twelve-variable equation)**.** *Every canonical native refutation witness extends to an integer zero of* P_{12}*.*

Proof. Retain its front-end coordinates and its roots d, τ, and first choose t with $O > 0$. Lemma 6.1 gives $s > 0$ and an integer g with $g^2 = (3O - 4)s^2 + 1$. Apply Theorem 6.2 with $N = s$. It gives a fixed I divisible by sc^2; put $r = I/(sc^2)$. Write $\gamma = (d - b)/H \in \mathbb{Z}$.

By Lemma 3.1, choose $a_0 \pmod I$ with $\Pi(\gamma + a_0) \equiv 0 \pmod I$. Use the interpolation theorem to choose the same auxiliary T in that residue. Then

$$\Theta(H, d - b, T) = H^5\Pi(\gamma + T)$$

is divisible by $S = IH^5$. The auxiliary root gives $a_1 = (Q-1)e$ with $a_1^2 = D_{\mathrm{a}}$. All five radicals in (4.4) now have integral roots. Choose

$$n = \frac{\mathcal{A} + \mathcal{B}d}{S} + W_1 a_1 + W_2 f + W_3 \tau + W_4 g.$$

Every quantity on the right has already been chosen, and the quotient is integral. It makes one norm factor vanish. There is no circular selection of I, T, n, and no extra divisibility promise on the canonical native data. □

7 Exact degree and the literal norm circuit

For three radicals use the eight-factor norm

$$\mathcal{F}_8(x; A, B, C) = \left((x^2 - A - B - C)^2 - 4(AB + AC + BC)\right)^2 - 64ABCx^2. \tag{7.1}$$

Compute $\mathcal{F}_8(x - u - v; A, B, C)$ in the rank-four algebra with $u^2 = D$, $v^2 = E$, using the basis $1, u, v, uv$. If the resulting vector is (f_0, f_1, f_2, f_3), set

$$\begin{aligned} g_0 &= f_0^2 + Ef_2^2 - D(f_1^2 + Ef_3^2), \\ g_1 &= 2(f_0 f_2 - D f_1 f_3), \\ \mathcal{F}_{32}(x; A, B, C, D, E) &= g_0^2 - Eg_1^2. \end{aligned}$$

Every operation is polynomial. The archive retains the small inherited norm implementation and checks it against an independent formal conjugate product. The twelve-variable graph specializes the cleared norm routine at denominator 1; no denominator leaf remains.

The important front-end degrees of P_{12} are

register	a	H	Δ	I	D_{f}	Q	V	D_{a}	O	D_{g}
deg	6	6	12	4	20	32	25	114	44	46.

Also $\deg \mathcal{A} = 35$, $\deg \mathcal{B} = 28$, $\deg S = 34$ and $\deg Z = 35$. The bound degrees are $4, 58, 12, 20$ for $B_0, B_{\mathrm{a}}, B_{\mathrm{f}}, B_\tau$, respectively. Thus $\deg(W_1, W_2, W_3, W_4) = (4, 62, 74, 94)$. The five radicands have respective degrees

$$70, \quad 190, \quad 212, \quad 250, \quad 302.$$

The last is uniquely largest, and $\deg Z = 35 < 151$. The unique highest contribution in the norm is therefore T_4^{16}, of degree 4832.

For a specialization-proof nonzero coefficient put

$$q_0 = 6\beta j, \qquad Y_0 = 2vq_0^3, \qquad a_0 = 24wvq_0^4, \qquad I_0 = src^2.$$

The leading form of the guard is $(D_{\mathrm{g}})_{46} = -3s^2h^4a_0^4Y_0^4$. Together with the weight definitions this gives

$$(P_{12})_{4832} = 2^{704}3^{16}a_0^{544}Y_0^{128}I_0^{160}c^{64}s^{32}h^{64}(T^2 + c^2)^{64}(s^2 + r^2)^{32}T^{64}.$$

Selecting T^{128} from the first binomial and r^{64} from the second gives

$$[j^{2560}w^{544}v^{672}h^{64}s^{192}r^{224}c^{384}T^{192}]P_{12} = 2^{5024}3^{3120}\beta^{2560} > 0. \tag{7.2}$$

The variable exponents sum to 4832. This is an exact degree certificate on the actual fixed coefficient tuple.

7.1 A shorter circuit at degree 16000

There is a useful operation–degree tradeoff. Use the same front end, $S, Z, \mathcal{A}, \mathcal{B}$, but set

$$B_*^{\mathrm{short}} = 1 + \mathcal{B}^2 D_{\mathrm{m}} + D_\tau + D_{\mathrm{f}} + D_{\mathrm{a}} + D_{\mathrm{g}}^2, \qquad \mathcal{W}^{\mathrm{short}} = 2B_*^{\mathrm{short}} + 1.$$

and replace the five radicands by

$$\mathcal{B}^2 D_{\mathrm{m}}, \quad S^2(\mathcal{W}^{\mathrm{short}})^2 D_{\mathrm{a}}, \quad S^2(\mathcal{W}^{\mathrm{short}})^4 D_{\mathrm{g}}, \quad S^2(\mathcal{W}^{\mathrm{short}})^6 D_\tau, \quad S^2(\mathcal{W}^{\mathrm{short}})^8 D_{\mathrm{f}}.$$

Use these in the same $\mathcal{F}_{32}$ norm centered at Z. The same proof works; even before the guard has been recovered, this bounds all five unweighted absolute magnitudes once $S \neq 0$. Its literal graph costs 223 binary operations. Here $\deg \mathcal{W}^{\mathrm{short}} = 114$, and the five radicand degrees are $70, 410, 570, 786, 1000$. The exact degree is 16000. Writing

$$R_0 = -q_0^3(12wC - q_0 z), \qquad b_0' = 4I_0 T - R_0,$$

its highest homogeneous part is

$$2^{448} a_0^{1728} I_0^{1088} (b_0')^{256}.$$

In particular the coefficient of $j^{6912} w^{1728} v^{1728} s^{1344} r^{1344} c^{2688} T^{256}$ is $2^{13056} 3^{8640} \beta^{6912} > 0$. Both versions have exactly the same twelve integer coordinates, the same six coefficient numerals, and the same solvability equivalence. The archive supplies both literal graphs.

8 Thirteen integers: trace before separation

The companion equation has thirteen unrestricted integer coordinates, exact degree 1072, and 190 live binary operations. It improves the inspected CRT companion of [2], which has degree 1088 and 195 operations, while preserving its complete fixed coefficient tuple. The key step recovers a divisibility condition from a norm zero before requiring individual radicals to be rational. Consequently the original large-scale separation bounds become available earlier, and two scaling devices can be removed.

8.1 Complete specification

Use the six constants (2.1) and exactly the coordinates

$$\boxed{j, C, z, h, r, v, w, T, c, n, t, p_2, p_3 \in \mathbb{Z}.} \tag{8.1}$$

The common front end is recomputed with $Y = qv$:

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W_0 &= 12w + 4,\\
X &= qW_0, & Y &= qv, & E &= XY,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
F &= (K_0 + W_0)C - z(q-1), & \alpha &= q - F - C, & &\\
R &= \big(q(q-F) - C\big)(q^2 - 1) + (M_C + qM_F)J, & & & &\\
k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta.
\end{aligned} \tag{8.2}$$

Let G, O have their exact definitions in (2.5), using these new values of F, q, η, ζ. Define

$$\begin{aligned} I &= (2r+1)c^2, & D_{\mathrm{f}} &= 1+\Delta I^2, & Q &= \Delta^2 I^2, \\ \delta &= Q-1, & V &= D_{\mathrm{f}}(cT-R)-c, & C_y &= QV^2-1, \\ L_\tau &= XY^2k, & D_\tau &= 1+L_\tau(L_\tau+k), & & \\ D_{\mathrm{m}} &= 1+\Delta c^2, & b &= X+ac, & & \\ A_g &= O-1-p_2^2-p_3(p_3+1), & & & & \\ M &= 2q^2, & U &= MH, & Z &= U(n+V)+Mb+H(v-q^2). \end{aligned} \tag{8.3}$$

In particular this section's I, V, A_g differ from the twelve-variable definitions. There is no multiplier v^2 in A_g and no factor $4H+1$ on the main radical. Using the polynomial (4.5), put

$$\boxed{P_{13}^{\mathrm{tr}} = \delta^{16}\mathcal{F}_{32}\left(Z; M^2D_{\mathrm{m}}, U^2D_{\mathrm{f}}, U^2D_\tau, U^2A_g, \frac{U^2C_y}{\delta}\right) = 0.} \tag{8.4}$$

The degree in the final radicand of $\mathcal{F}_{32}$ is at most sixteen, so the displayed denominator clears symbolically. The delivered arithmetic graph contains only integer leaves and binary $+, -, \times$; neither division nor radical extraction is an evaluator operation.

8.2 A congruence from the rational part

Lemma 8.1 (Rational-part divisibility)**.** *Let $a_i \in \mathbb{Q}$, possibly zero or negative, and choose roots $\alpha_i^2 = a_i$. Suppose every rational α_i is integral. If $Z, L_1, \ldots, L_m \in \mathbb{Z}$ and $Z = \sum_i L_i\alpha_i$, then $\gcd(L_1, \ldots, L_m)$ divides Z.*

Proof. In the finite multiquadratic splitting field $K = \mathbb{Q}(\alpha_1, \ldots, \alpha_m)$ apply $\pi(x) = \mathrm{Tr}_{K/\mathbb{Q}}(x)/[K : \mathbb{Q}]$. A rational root is fixed and contributes its integer value. An irrational root has minimal polynomial $x^2 - a_i$, of trace zero; trace transitivity therefore gives $\pi(\alpha_i) = 0$. Thus $Z = \sum_i L_i\pi(\alpha_i)$ is an integer linear combination of the L_i. Imaginary roots also have rational part zero. Independence of square classes and separation of magnitudes are unnecessary. □

Here the four radicands $D_{\mathrm{m}}, D_{\mathrm{f}}, D_\tau, A_g$ are integers. For the remaining rational radicand $D_y = C_y/\delta$, a rational root is integral by Lemma 5.1 when $Q > 1$; when $Q = 0$ it equals ± 1. The following checks establish that these cases exhaust every integer assignment, before imposing the equation.

Always q is even, $q \equiv 2 \pmod 3$, $|q| \geq 16$, and $|X| \geq 64$. At $v = 0$ we have $Y = a = 0$, $H = 3$, and $\Delta = 3$. Otherwise $|Y| \geq 16$ and $|a| \geq 63 \cdot 16$. Hence in both cases $H \neq 0$ is odd and $\Delta \geq 3$ is positive and odd. The radicands $D_{\mathrm{f}}, D_{\mathrm{m}}$ are at least one. Writing $u = XY^2 \in \mathbb{Z}$ gives $D_\tau = 1 + u(u+1)k^2 \geq 1$.

At $c = 0$, $Q = 0, \delta = -1, C_y = -1$, and $D_y = 1$. At $c \neq 0$, $2r+1 \neq 0$ gives $Q \geq 9$ and $D_{\mathrm{f}} > |c|+1$. For the integer $b_1 = cT - R$, either $b_1 = 0$ and $|V| = |c| \geq 1$, or $|V| \geq D_{\mathrm{f}} - |c| > 1$. Thus

$$\delta \neq 0, \qquad D_y = V^2 + \frac{V^2-1}{Q-1} \geq 1 \quad (c \neq 0). \tag{8.5}$$

All four non-guard radicands are positive, including on the exceptional $c = 0$ and $v = 0$ slices.

At a zero of (8.4), one conjugate factor vanishes. Absorb its signs into roots d, f, τ, g, y to obtain

$$Z = Md + U(f+\tau+g+y), \qquad (d^2, f^2, \tau^2, g^2, y^2) = (D_{\mathrm{m}}, D_{\mathrm{f}}, D_\tau, A_g, D_y). \tag{8.6}$$

Lemma 8.1 gives $M \mid Z$ immediately. This does not assume that all five roots are rational or that $A_g \geq 0$. Since

$$Z = M\big(H(n+V)+b\big) + H(v-q^2),$$

we have $M \mid H(v-q^2)$. But $Y = qv$ implies $H \equiv 3 \pmod q$, and $3 \nmid q$ with H odd gives

$$\gcd(H, M) = 1, \qquad v = q^2(2v_0+1), \qquad Y = (2v_0+1)q^3, \quad v_0 = \frac{v-q^2}{2q^2} \in \mathbb{Z}. \tag{8.7}$$

Thus $v, Y \neq 0$ and $|Y| \geq |q|^3 \geq 4096$ are established before squarehood or Pell-index recovery. This early recovery is the new step.

8.3 Unscaled guard, separation, and restoration

If $A_g < 0$, its radical in (8.6) would be the only nonzero imaginary term; therefore $A_g \geq 0$. Consequently $O = A_g + 1 + p_2^2 + p_3(p_3+1) > 0$, and (2.6) gives $G > 0$ and $\eta\zeta > 0$. This also excludes $c = 0$, since there $\eta\zeta = -Y(Y+1)k^2 \leq 0$.

Put $x = |X| \geq 64$, $m = |Y| \geq 4096$, $K = |k|$, and $C_0 = |c|$. The strip guard gives $K \geq 2$ and

$$K(m-1) < C_0 < K(m+1). \tag{8.8}$$

Denote positive roots by d_+, f_+, τ_+, y_+, and let $g_+ = \sqrt{A_g} \geq 0$. With $a_* = |a| \geq (x-1)m$, the bounds $\sqrt{\Delta} > a_* - 3 > xm/2$ yield

$$\begin{gathered} f_+ > \frac{xm}{2}C_0^2 > 3C_0^2, \qquad K(xm^2-1) < \tau_+ < K(xm^2+1), \\ \tau_+ > 2C_0, \qquad f_+ > 2\tau_+, \qquad Q > D_{\mathrm{f}} > D_\tau > 1, \quad Q > C_0^2, \\ \frac{C_0}{5} < \frac{d_+}{|H|} < \frac{C_0}{2}. \end{gathered} \tag{8.9}$$

For example, $f_+ > 2\tau_+$ follows from $mK(m-1)^2 > 4(m^2 + 1/x)$. The last comparison uses $a_* - 3 < \sqrt{\Delta} < a_* + 3$ and $4a_* - 3 \leq |H| \leq 4a_* + 3$. The determinant identity

$$12G = 4q^2 - \big(3(U_c + Z_c) - 4q\big)^2 - 3(U_c - Z_c)^2$$

gives $G < q^2$. Since $\eta\zeta \leq k^2/4$, the unscaled guard gives

$$g_+ < \frac{|q|K}{2} < \frac{C_0}{10}. \tag{8.10}$$

Here the final inequality uses the already recovered $m \geq |q|^3$.

If $b_1 = cT - R = 0$, then $V = -c$ and $C_0 < y_+ < C_0 + 1$. Otherwise $|V| \geq f_+^2 - C_0 > 2f_+$ and $y_+ > |V| > 2f_+$. After omitting a zero g_+, the following orders are strictly superincreasing:

$$\begin{aligned} b_1 = 0: &\quad g_+,\ d_+/|H|,\ y_+,\ \tau_+,\ f_+, \\ b_1 \neq 0: &\quad g_+,\ d_+/|H|,\ \tau_+,\ f_+,\ y_+. \end{aligned} \tag{8.11}$$

The two smallest sum to less than $0.6C_0$; in the first case the three smallest sum to less than $1.6C_0 + 1 < 2C_0 < \tau_+$. The other comparisons follow from (8.9). Divide (8.6) by U. Its main-root coefficient is $M/U = 1/H$, so Lemma 5.2 applies to precisely these magnitudes. All roots are rational; the integer radicands make d, f, τ, g integral, and Lemma 5.1 makes y integral.

Reducing the integral relation (8.6) modulo H and using (8.7) recovers $\gamma = (d-b)/H \in \mathbb{Z}$. Thus we have the exact normalized kernel

$$\begin{gathered} f^2 = D_{\mathrm{f}}, \quad \tau^2 = D_\tau, \quad QV^2 - (Q-1)y^2 = 1, \\ (b + H\gamma)^2 = D_{\mathrm{m}}, \quad A_g = g^2, \quad Y = (2v_0+1)q^3. \end{gathered} \tag{8.12}$$

It is the incoming unscaled normalized integer kernel of [2], with its old scale coordinate v_0 and its guard square coordinate replaced by g. Its transport substitution and entire fixed tuple are unchanged.

Conversely, take any solution of that kernel, writing its guard root as p_1 and its main root as $d = b + H\gamma$. Set

$$v = q^2(2v_0 + 1), \qquad n = \gamma + f + \tau + p_1 + y - V - v_0. \tag{8.13}$$

Every other supplied coordinate is retained. These integer choices give (8.6), so one norm factor vanishes. The two directions prove an existence equivalence with the unchanged kernel. Its ordinary native/K215 interpretation is the inherited dependency described in Section 9; it is not supplied by a finite norm test.

8.4 Exact degree and complete arithmetic cost

Removing v^2 from the old guard saves two gates; removing its optimized main scale $4U + M$ saves two more. The established factorization $D_\tau = 1 + L_\tau(L_\tau + k)$ saves one further gate. The complete new count is 118 multiplications, 40 additions and 32 subtractions. All thirteen variables, all six ports, and every one of the 190 gates are live. This includes the whole cleared norm.

For the degree calculation write

$$d = M^2 D_{\mathrm{m}}, \quad e = U^2 A_g, \quad A' = U^2 D_{\mathrm{f}}, \quad B' = U^2 D_\tau, \quad C' = U^2 C_y,$$

and $s_0 = Un + Mb + H(v - q^2)$, so $Z = UV + s_0$. Their degrees are $14, 40, 26, 34, 72$, while $\deg U = 6$, $\deg V = 19$, $\deg \delta = 22$, and $\deg s_0 = 7$. The first scalar component of the biquadratic cleared three-root computation is exactly

$$J_0 = \delta(s_0^2 + 2UV s_0 + d + e - A' - B') - U^2(V^2 - 1).$$

Giving $\sqrt{d}, \sqrt{e}$ weights $7, 20$, exact sparse expansion gives the successive four-component weighted degrees

$$(62, 54, 67, 49), \quad (134, 116, 129, 121), \quad (268, 250, 263, 255). \tag{8.14}$$

The unique top scalar is $16\delta^4 U^4 V^4 e^2$ of degree 268. The two remaining quadratic norms square it twice, giving degree 1072.

For explicit attainment on the actual coefficient slice put

$$q_0 = 6\beta j, \quad a_0 = 12wvq_0^2, \quad i_0 = 2r, \quad R_0 = q_0^4 z - 12q_0^3 wC, \quad \eta_0 = -q_0 v(R_0 + ha_0).$$

The full leading form is

$$(P_{13}^{\mathrm{tr}})_{1072} = 2^{112} q_0^{64} a_0^{128} i_0^{64} c^{128} R_0^{16} \eta_0^{32}. \tag{8.15}$$

In particular,

$$[j^{480} w^{160} v^{192} h^{32} r^{64} c^{128} z^{16}] P_{13}^{\mathrm{tr}} = 2^{976} 3^{640} \beta^{480} > 0. \tag{8.16}$$

The exponent sum is 1072. The h^{32} and then z^{16} selections are unique, so relations among the other five constants cannot cancel this coefficient. The degree certificate checks these identities exactly; modular evaluations separately check transcription of the complete graph.

9 Consistency calibration and proof boundaries

Theorem 9.1 (Ordinary consistency equivalence)**.** *For the fixed tuple in Appendix A, both* P_{12} *and* P_{13}^{tr} *satisfy* (1.1)*, with respectively twelve and thirteen unrestricted integer coordinates.*

Proof. For P_{12}, Section 5 restores the main and first Pell roots, strict determinant and strip guards, a nonzero integer I, and both auxiliary congruences. Write $B_0 = |a+2| \geq 2$ and $c_0 = |c|$. The inherited signed first/main rank bootstrap gives $c_0 = \psi_{B_0}(p)$ with $p \geq 25$ [4, 5]. The normalized strong equation $f^2 - \Delta I^2 = 1$ directly gives $|I| = \psi_{B_0}(m)$ and $|f| = \chi_{B_0}(m)$ for some $m > 0$. Since $c_0^2 \mid |I|$, strong divisibility gives $p \mid m$. Write $m = pk$, and put $C_0 = \chi_{B_0}(p)$. Binomial expansion yields

$$\frac{\psi_{B_0}(pk)}{c_0} \equiv kC_0^{k-1} \pmod{c_0}.$$

The left side is divisible by c_0, whereas $\gcd(C_0, c_0) = 1$. Thus $c_0 \mid k$, and in particular $c_0 \mid m$ and $m > 2p$. These are precisely the strong-index inputs for the inherited signed auxiliary step-down. No generalized odd-multiplier lemma is required.

For that step-down put $\rho = \Delta I$. Then

$$\rho^2 \equiv -\Delta \pmod{|f|}, \qquad \rho^2 \equiv 0 \pmod{c_0}.$$

The exposed quotient equations also have integral polynomial witnesses:

$$\begin{aligned} o &= f(c - R + 4IT), \\ j_{\mathrm{aux}} &= \Delta s^2 r^2 c^3 (c - R) + 4src\, D_{\mathrm{f}} T, \\ V &= fo - c = cj_{\mathrm{aux}} - R. \end{aligned}$$

The auxiliary Pell index is odd because $\chi_{|\rho|}(2k) \equiv (-1)^k \pmod{|\rho|}$, independently of the parity of ρ. The signed rank and ratio argument now gives $p = R$, $2n_{\mathrm{first}} = R + 1$, and $X = \mathrm{sign}(c)2^R$. Since $X \equiv 2 \pmod 3$, it selects $c > 0$ and $X = 2^R$. The first-index sign identity gives $k > 0$, hence $Y > 0$. Finally the Mersenne-radix sign lemma excludes negative q. The post-power native decoder therefore applies directly [4, 5]. These retained proofs cover the signed stages before any positive-domain theorem is invoked.

Conversely, the inherited canonical native construction supplies (2.7), including its even A and $R \equiv 3 \pmod 4$. The interpolation theorem constructs a new strong and auxiliary witness without altering those native coordinates. The completeness proof then constructs the polynomial zero.

The trace companion has a direct two-way restoration to the incoming normalized integer kernel, as proved in its section. The inherited native-window/compiler theorem identifies the kernel's existence with blank halting of the fixed K215 machine, and the retained ordinary K215 calibration identifies that halting with existence of a finite refutation of the specified ZFC calculus. These are mathematical theorems used as lemmas, with their exact source versions and explanatory proofs retained. All new arguments use finite algebra, induction, elementary number theory, Pell recurrences, finite fields and finite coding, available in ZFC. Compose the existence equivalences and negate them. □

Three assertions must be distinguished. First, the new reductions have the ordinary unbounded proofs supplied here. Second, the consistency interpretation imports the named ordinary native and syntactic-calibration theorems; this report does not reprove the entire compiler from its local tile list. Third, executable audits certify finite identities, numeral bindings, degrees and graph transcription. Those audits are not proofs of the unbounded compiler theorem. The repository separately records that its complete native Lean calibration and encoded object-ZFC derivation remain unfinished [7, 8]. This package does not change that status.

The inherited native/compiler and K215 texts supply ordinary proofs, not conjectural placeholders. The dependency ledger identifies the exact lemmas being composed. A future proof-assistant formalization must cover that composition as well as the final polynomial algebra.

10 Reproduction, attribution and further research

The archive contains the three literal equations, builders, exact algebra checks, separate mathematical audits, a manifest, all six fixed numeral payloads, and their complete finite reconstruction. The retained `inherited/root_fusion` directory is an unchanged copy of the inspected incoming source package. Its licenses, provenance and proof boundaries remain attached. No existing repository files were overwritten or published by this delivery.

The top-level verification entry point regenerates the graphs in memory, checks the delivered equations and coefficients, runs independent transcription diagnostics and negative controls, and writes an explicit receipt. The precise reproduction commands and dependency requirements are in `README.md`. Python optimization must not remove acceptance checks. No verifier searches for a consistency witness, simulates an unbounded machine execution, or treats a bounded lack of zeros as evidence of consistency. The mathematical reviews are internal independent reviews of this research session, not external human peer review.

10.1 Further research questions

1. **Reduce the separating weights further.** Bounding square-root magnitudes instead of radicands already reduces the twelve-variable degree from 16000 to 4832, using homogeneity and distinct weights. Further source-specific bounds may improve the tradeoff again. Any such change must retain the order of the proof: exclude $S = 0$, recover positivity, and then separate arbitrary integer assignments.
2. **Fuse square conditions through proved coprimality.** The strong and auxiliary conditions might be combined into a single square if the multiplier is chosen to absorb precisely the obstructing prime factors. Any saving must include the resulting degree increase and must preserve full residue interpolation of the shared T.
3. **Use trace before separation elsewhere.** The trace lemma extracts congruences even when irrational radicals cancel. Search for other redundant guard multipliers and scaling weights in the existing nine-positive and ten-integer circuits. Rational radicands require a separate integrality theorem; monicity cannot be assumed.
4. **Find smaller real-quadratic-safe mergers.** The monotone quintic supplies intersectivity, real-quadratic nonvanishing and a nonzero odd coefficient simultaneously. Can another polynomial reduce the degree or number of operations while retaining all three properties? An ordinary rational-root-free polynomial is insufficient here.
5. **Remove another witness through a proved shared family.** Setting $I = sc^2$ would erase r, but would require simultaneous Pell solvability with a prescribed positivity witness. Neither this report nor the finite checks establish that assertion. A theorem furnishing such a family would be a substantive further variable reduction.
6. **Complete the native formal certificate.** The most valuable formalization target is the shared source-to-refutation calibration and its internal-ZFC export. Certifying only the small final norm identities does not close that semantic boundary.
7. **Compare alternative source encodings on equal terms.** The direct numerical-calculus quartics already in the incoming collection avoid K215 and Jones, at a cost in variables. The norm and interpolation lemmas here are not intrinsically tied to K215; applying them to an alternative calibrated source requires an actual complete source packet and new counts, not an unspecified universal coefficient.

A Complete bounded coefficient recipe

The following is the full source-34 finite arithmetic recipe [3], reproduced from the incoming root-fusion report with its licenses retained. All ranges are bounded constants. The ordered 943 tile records $(x_{i0}, x_{i1}, x_{i2}, x_{i3})$, in west, north, east, south order, are supplied in `inherited/root_fusion/coefficients/recipe/wang943_tiles.json`. Their ordering is part of the definition: selector 0 is Start and selector 1 is Forbidden. The supplied `k215.tm` contains all 430 working actions from which the source builder reconstructs the whole ordered list. Its SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090`.

Thus this finite table is delivered data, not an unspecified machine or a free coefficient parameter. Put

$$b = 121, \quad L_0 = 8801, \quad d = bL_0 = 1064921, \quad \rho = 2^{121}, \quad B_0 = 2^{1064921}.$$

For $0 \le s < 4$, $0 \le j < 10$, define

$$e_s(j) = o_s + 2j, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960),$$

and $w_j = 2^{10+2j}$ for $0 \le j < 44$. Let

$$\mathcal{E} = \{0, \dots, 942\} \cup \{e_s(j) : 0 \le s < 4,\ 0 \le j < 10\} \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.$$

It has 989 elements and maximum 3296. For each tile $0 \le i < 943$, set

$$c_i = 1 + \sum_{s=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{is} + 512) w_{10s+j} + \sum_{j=40}^{43} w_j.$$

Here $\mathrm{bit}_j(m) = \lfloor m/2^j \rfloor \bmod 2$ is used only to define the finitely many fixed integer coefficients. It is not an operation on a quantified Diophantine variable. The other coefficients are

$$c_{e_s(j)} = w_{10s+j}, \quad (c_{1095}, c_{1096}, c_{2195}, c_{3296}) = (w_{40}, w_{41}, w_{42}, w_{43}), \quad c_{1063} = c_{1064} = 0.$$

Define

$$\mu = 1022 + \sum_{j=0}^{43} w_j,$$

$$Z_{\mathrm{pad}} = (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799},$$

$$D_C = \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e (\rho^{4393-e} + \rho^{5495-e}),$$

$$D_R = \rho,$$

$$M_C = B_0 - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{1063},$$

$$M_{F,\mathrm{native}} = 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) + \rho^{2195} + \rho^{3296} + Z_{\mathrm{pad}}.$$

The six actual fixed ports are precisely

$$\beta = B_0 - 1, \quad K_0 = D_C + B_0 D_R, \quad M_C, \quad M_F = M_{F,\mathrm{native}} + B_0 - 1, \quad A_{\mathrm{cone}} = 2^{1113}, \quad B_{\mathrm{cone}} = 2^{666104}.$$

Fixed numeral	Bit length	Minimal unsigned bytes
β	1 064 921	133 116
K_0	1 065 043	133 131
M_C	1 064 921	133 116
M_F	1 064 922	133 116
A_{cone}	1 114	140
B_{cone}	666 105	83 264
Total		615 883

Table 2: Exact fixed payload sizes before compression. The archive stores deterministic gzip encodings of the minimal unsigned little-endian bytes and binds their raw hashes.

All displayed exponents are fixed integers. In particular, M_F already includes its shift; shifting it again would change the equation and invalidate the binding.

The source proof verifies the whole coefficient interface, not just these sizes. It establishes local tile decoding, carry-free geometry, whole-cell rotation, mask populations, positive canonical slack, and an explicit rank controller. In particular

$$\operatorname{pc}(M_C) + \operatorname{pc}(M_{F,\text{native}}) = d, \quad M_C \equiv 2 \pmod 4, \quad M_{F,\text{native}} \equiv 4 \pmod 8.$$

The controller uses $d = 11^2 \cdot 13 \cdot 677$, $\lambda = 2341$, and a proved Boolean subset coverage lemma; it does not assume that d is a power of five. Its construction supplies odd canonical exponents, odd cubic scale, and the canonical parity promises in (2.7). The complete inherited proof and literal builder are included, with their internal source pins and licenses preserved.

References

[1] V. Reshetnikov, *ProveIt*, reviewed commit `04a1d79222d55cabe4ba52897863fc33baa8c268`. https://github.com/VladimirReshetnikov/ProveIt/tree/04a1d79222d55cabe4ba52897863fc33baa8c268/Computability/DiophantineConsistency.

[2] *Thirteen Integer Variables through a Single Five-Radical Norm*, incoming package `ConZFC_13_Integer_Norms_2026-10-24.zip`, in [1]. Complete package retained under `inherited/root_fusion`.

[3] ProveIt source-34 finite coefficient recipe and ordinary compiler proof, retained in [2], `provenance/upstream_ordinary/source34_coefficients.md` and `coefficients/recipe/`.

[4] ProveIt ordinary and signed Pell/native proofs, retained in [2], `provenance/upstream_ordinary/ordinary36.md`, `provenance/RESIDUE_AND_DETERMINANT.md`, and `provenance/NORMALIZED_NORMS_PROOF.md`.

[5] ProveIt, *Joint guards and signed residue orientation*, incoming `ZFC_Diophantine_Joint_Guards_2026-10-24.zip`. The two relevant proofs and their notices are retained in `provenance/signed_guards/`.

[6] ProveIt ordinary K215 consistency calibration, retained in [2], `provenance/upstream_ordinary/k215_ordinary.md`.

[7] ProveIt, *A Diophantine equation for syntactic consistency of ZFC*, active research README at the reviewed commit.

[8] ProveIt, *Object-ZFC formalization status*, as retained in the incoming package's `provenance/CALIBRATION_STATUS.md` and pinned source ledger.

[9] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, arXiv:2609.01594v2, 2026. https://arxiv.org/abs/2609.01594.

[10] C. Sanna, *The p-adic valuation of Lucas sequences*, Fibonacci Quarterly **54** (2016), 118–124. The valuation results are also explicitly stated and applied in [9], Lemmas 2.3–2.4.